

Aspirin: A History

Medicines made from willow and other salicylate-rich plants appear in clay tablets from ancient Sumer as well as the Ebers Papyrus from ancient Egypt. Hippocrates referred to the use of salicylic tea to reduce fevers around 400 BC, and willow bark preparations were part of the pharmacopoeia of Western medicine in classical antiquity and the Middle Ages. Willow bark extract became recognized for its specific effects on fever, pain, and inflammation in the mid-eighteenth century after the Rev Edward Stone of Chipping Norton, Oxfordshire, noticed that the bitter taste of willow bark resembled the taste of the bark of the cinchona tree, known as "Peruvian bark", which was used successfully in Peru to treat a variety of ailments. Stone experimented with preparations of powdered willow bark on people in Chipping Norton for five years and found it to be as effective as Peruvian bark and a cheaper domestic version. In 1763, he sent a report of his findings to the Royal Society in London. By the nineteenth century, pharmacists were experimenting with and prescribing a variety of chemicals related to salicylic acid, the active component of willow extract.

In 1853, the chemist Charles Frederic Gerhardt treated sodium salicylate with acetyl chloride to produce acetylsalicylic acid for the first time; in the second half of the 19th century, other academic chemists established the compound's chemical structure and devised more efficient methods of synthesis. In 1897, scientists at the drug and dye firm Bayer began investigating acetylsalicylic acid as a less-irritating replacement for standard common salicylate medicines, and identified a new way to synthesize it. That year, Felix Hoffmann (or Arthur Eichengrün) of Bayer was the first to produce acetylsalicylic acid in a pure, stable form.

Salicylic acid had been extracted in 1839 from the herb meadowsweet, whose German name, *Spirsaure*, was the basis for naming the newly synthesized drug, which, by 1899, Bayer was selling globally. The word Aspirin was Bayer's brand name, rather than the generic name of the drug; however, Bayer's rights to the trademark were lost or sold in many countries. Aspirin's popularity grew over the first half of the 20th century, leading to fierce competition with the proliferation of aspirin brands and products.

Aspirin's popularity declined after the development of acetaminophen/paracetamol in 1956 and ibuprofen in 1962. In the 1960s and 1970s, John Vane and others discovered the basic mechanism of aspirin's effects, while clinical trials and other studies from the 1960s to the 1980s established aspirin's efficacy as an anti-clotting agent that reduces the risk of clotting diseases. The initial large studies on the use of low-dose aspirin to prevent heart attacks that were published in the 1970s and 1980s helped spur reform in clinical research ethics and guidelines for human subject research and US federal law, and are often cited as examples of clinical trials that included only men, but from which people drew general conclusions that did not hold true for women.

Bayer lost its trademark for aspirin in the United States and some other countries in actions taken between 1918 and 1921, because it had failed to use the name for its own product correctly and had for years allowed the use of "Aspirin" by other manufacturers without defending the intellectual

property rights. Aspirin, with a capital "A", remains a registered trademark of Bayer in Germany, Canada, Mexico, and over 80 other countries.